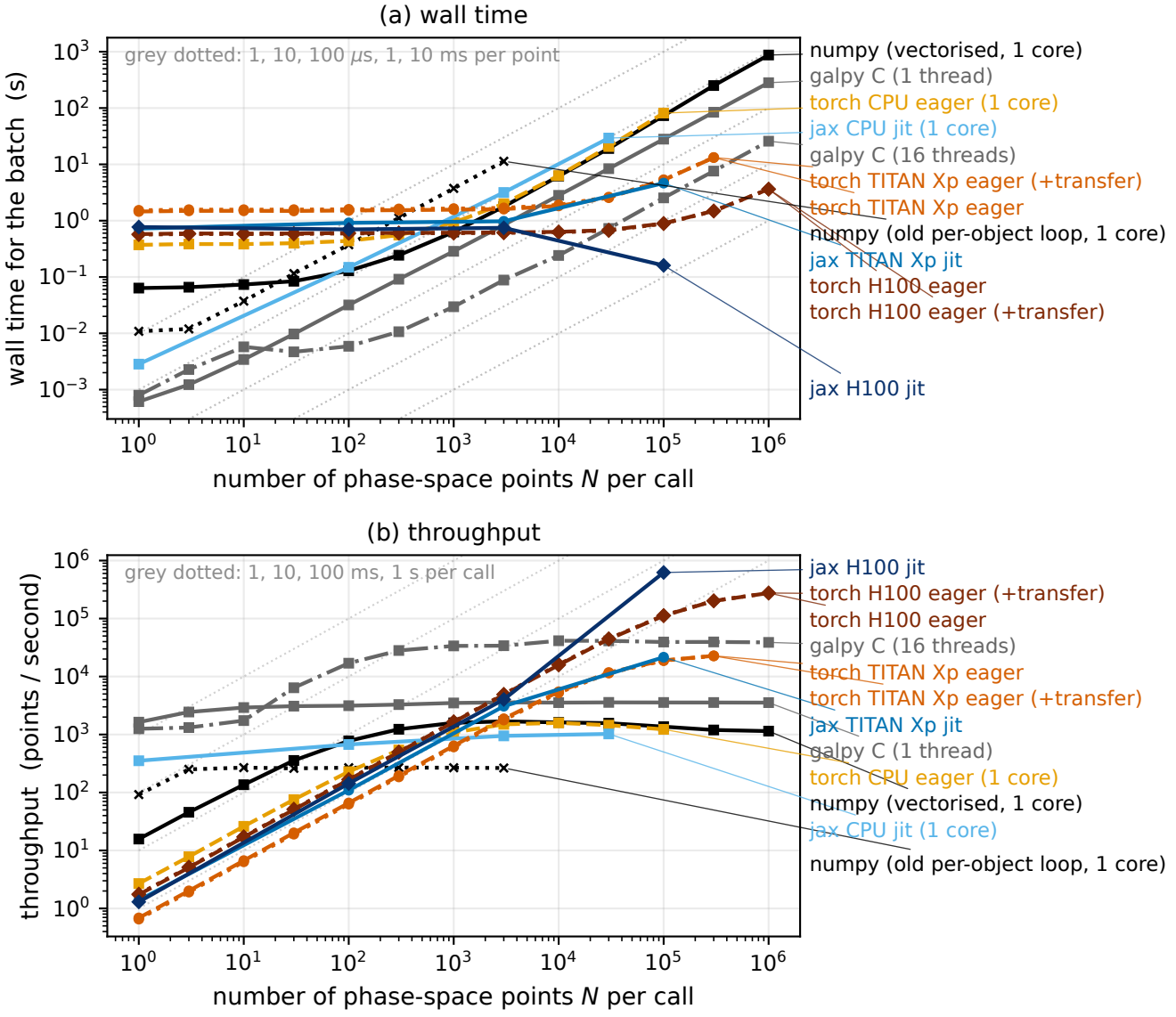
(a) wall time
grey dotted: 1, 10, 100 μs, 1, 10 ms per point
wall time for the batch (s)
number of phase-space points N per call
numpy (vectorised, 1 core)
galpy C (1 thread)
torch CPU eager (1 core)
jax CPU jit (1 core)
galpy C (16 threads)
torch TITAN Xp eager (+transfer)
torch TITAN Xp eager
numpy (old per-object loop, 1 core)
jax TITAN Xp jit
torch H100 eager
torch H100 eager (+transfer)
jax H100 jit
(b) throughput
grey dotted: 1, 10, 100 ms, 1 s per call
throughput (points / second)
number of phase-space points N per call
jax H100 jit
torch H100 eager (+transfer)
torch H100 eager
galpy C (16 threads)
torch TITAN Xp eager
torch TITAN Xp eager (+transfer)
jax TITAN Xp jit
galpy C (1 thread)
torch CPU eager (1 core)
numpy (vectorised, 1 core)
jax CPU jit (1 core)
numpy (old per-object loop, 1 core)